

AI Website Builder

Deployment Guide

Step-by-step instructions to deploy your app to production

What You Need Before Starting

- A GitHub account (github.com) — free
- A Firebase account (firebase.google.com) — free
- A Vercel account (vercel.com) — free
- A Google AI Studio API key (aistudio.google.com) — free tier available
- The project source code (the zip file you received)

Estimated time: 30–45 minutes. No coding experience required — just follow each step carefully.

Part 1 — Upload Code to GitHub

GitHub stores your source code and connects to Vercel for automatic deployments.

1

Create a GitHub account (if you don't have one)

Go to github.com and click Sign Up. Follow the prompts to create a free account. Verify your email address before continuing.

2

Create a new repository

After logging in, click the green New button (top-left area). Name your repository something like `ai-website-builder`. Set it to Private. Leave all other options at their defaults. Click Create repository.

3

Upload the project files

On the next page, click the link that says uploading an existing file. Unzip the project zip file you received on your computer. Drag ALL the unzipped files and folders into the GitHub upload area. Wait for all files to finish uploading, then scroll down and click Commit changes.

Tip: Make sure you see files like `vercel.json`, `README.md`, the `api/` folder, and the `frontend/` folder in your repository before moving on.

Part 2 — Set Up Firebase (Database)

Firebase provides the Firestore database that stores your website generation history.

1

Go to the Firebase Console

Visit console.firebase.google.com and sign in with your Google account. Click Add project.

2

Create a new project

Enter a project name such as ai-website-builder. On the next screen, you can disable Google Analytics (toggle it off) — it is not needed. Click Create project and wait about 30 seconds for it to finish.

3

Enable Firestore Database

In the left sidebar, click Build, then Firestore Database. Click Create database. Choose Start in production mode. Select a location closest to you (e.g. us-east1 for US East). Click Enable and wait for the database to be created.

4

Set Firestore security rules

Click the Rules tab at the top of the Firestore page. Delete all existing text in the editor. Copy and paste the following rules exactly:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
  match /sites/{siteId} {
    allow read: if true;
    allow create: if request.resource.data.keys().hasAll(['prompt', 'code', 'createdAt'])
    && request.resource.data.code.size() < 900000;
    allow update, delete: if false;
  }
}
}
```

Click Publish to save the rules.

5

Get your Firebase web app credentials

In the left sidebar, click the gear icon next to Project Overview, then Project settings. Scroll down to the Your apps section. Click the (web) icon to add a web app. Give it any nickname (e.g. website-builder-web). Click Register app. You will see a block of code containing your configuration values — keep this page open, you will need these values in Part 4.

Important: Write down or keep these six values handy — `apiKey`, `authDomain`, `projectId`, `storageBucket`, `messagingSenderId`, `appId`. You will enter them in Vercel in Part 4.

Part 3 — Deploy to Vercel

Vercel hosts your application and runs the secure serverless function that calls the Gemini AI — keeping your API key safely on the server.

1

Create a Vercel account

Go to vercel.com and click Sign Up. Choose Continue with GitHub and authorize Vercel to access your GitHub account. This links the two accounts automatically.

2

Import your GitHub repository

On the Vercel dashboard, click Add New, then Project. You will see a list of your GitHub repositories. Find ai-website-builder (or whatever you named it) and click Import.

3

Configure the build settings

Vercel should auto-detect the settings from `vercel.json`. If it asks you to configure manually, set: Framework Preset = Vite, Root Directory = . (leave blank / use repository root), Build Command = `cd frontend && npm install && npm run build`, Output Directory = `frontend/dist`.

Do NOT click Deploy yet — you must add environment variables first (next step).

4

Add environment variables

On the same configuration page, expand the Environment Variables section. You need to add SEVEN variables. Click Add for each one:

Variable Name	Where to Find It
GEMINI_API_KEY	From aistudio.google.com (see Part 3.5)
VITE_FIREBASE_API_KEY	Firebase config: <code>apiKey</code> value
VITE_FIREBASE_AUTH_DOMAIN	Firebase config: <code>authDomain</code> value
VITE_FIREBASE_PROJECT_ID	Firebase config: <code>projectId</code> value
VITE_FIREBASE_STORAGE_BUCKET	Firebase config: <code>storageBucket</code> value
VITE_FIREBASE_MESSAGING_SENDER_ID	Firebase config: <code>messagingSenderId</code> value
VITE_FIREBASE_APP_ID	Firebase config: <code>appId</code> value

5

Get your Gemini API key

Open a new browser tab and go to aistudio.google.com. Sign in with your Google account. Click Get API key in the top navigation. Click Create API key, then Create API key in new project. Copy the key shown (it starts with Alza...). Go back to Vercel and paste it as the value for GEMINI_API_KEY.

Security reminder: GEMINI_API_KEY does NOT start with VITE_ — this is intentional. Variables without VITE_ are server-side only and are never sent to the browser.

6

Deploy the application

Click Deploy. Vercel will start building your project — this takes 1–3 minutes. You will see a progress bar. When it finishes, you will see a Congratulations screen with a live URL like <https://ai-website-builder-xyz.vercel.app>. Click that URL to open your app!

Part 4 — Test Your Deployment

Before sharing your app, verify everything works correctly.

1

Test AI generation

Open your Vercel URL. In the prompt box on the left, type a simple description: A simple landing page for a coffee shop with a hero image and menu section. Click Generate Website. After 15–30 seconds a website should appear in the preview area.

2

Test responsive preview

After a site is generated, click the Mobile button in the toolbar. The preview should resize to a phone-sized frame. Click Desktop to switch back.

3

Test image upload

Click Add Image in the left panel and select a photo from your computer. In your prompt, mention 'use my uploaded image for the hero'. Click Generate Website — your image should appear in the result.

4

Test History

Click the History tab in the top right. You should see your generated sites listed. Click one to reload it. If history appears, Firestore is working correctly.

5

Test Refine Design

With a generated site showing, change your prompt to: Make the color scheme dark with purple accents. Click Refine Design. A new version incorporating your changes should appear.

Part 5 — Troubleshooting

Problem	Likely Cause	Fix
Build fails on Vercel	Missing environment variables	Go to Vercel > Settings > Environment Variables and make sure all 7 variables are set.
Generation returns an error	Invalid Gemini API key	Go to aistudio.google.com and create a new API key. Update GEMINI_API_KEY in your environment variables.
History tab is empty	Firestore not configured or rules are too strict	Double-check your Firebase VITE_ environment variables in Vercel. Make sure you have the correct Firestore rules.
Firestore permission denied	Firestore security rules too strict	Re-paste the security rules from Part 2, Step 4 and click Publish again.
App loads but preview is blank	AI returned empty response	Try a more detailed prompt. If it persists, check your Gemini API key quota at aistudio.google.com .
Images not showing in preview	Placeholder not used in prompt	Mention USER_IMAGE_PLACEHOLDER_1 in your prompt, e.g. 'use USER_IMAGE_PLACEHOLDER_1 as the user's profile picture'.

Still stuck? Check the Vercel function logs: go to your Vercel project dashboard, click the Deployments tab, click your latest deployment, then click Functions. The generate function logs will show any server-side errors.

Part 6 — Making Future Updates

If you ever receive an updated version of the code:

1

Upload new files to GitHub

Go to your GitHub repository. Navigate into the folder you want to update. Click the pencil (edit) icon on individual files to edit them, or drag new files into the repository using Add file > Upload files.

2

Vercel auto-deploys

Every time you push changes to GitHub, Vercel automatically rebuilds and redeploys your app within 1–2 minutes. No manual action needed.

3

Check the deployment

Visit your Vercel dashboard to see the build status. A green checkmark means the deployment succeeded.

Summary of Key URLs

Service	URL	What You Do There
GitHub	github.com	Store and update your code
Firebase	console.firebase.google.com	Manage your Firestore database

Vercel	vercel.com	Host your app, manage env variables
Google AI	aistudio.google.com	Manage your Gemini API key
Your App	your-app.vercel.app	The live website builder

Congratulations — your AI Website Builder is live! ■ Share your Vercel URL with anyone who wants to generate beautiful websites with AI.